ROBOTICS

OVERVIEW

Applying leadership and 21st century skills, student teams design, build, program, a mobile robotic vehicle to solve the annual challenge. Participants must engineer their entry according to specifications, conduct testing, and demonstrate problem-solving. The annual theme will be posted on the TSA website under *Themes & Problems*.

ELIGIBILITY

Two (2) teams of two to six (2-6) members per state may participate.

TIME LIMITS

A. Ten (10) minutes running clock to complete the challenge.

B. Ten (10) minutes for semifinalist interview

ATTIRE

TSA competition attire is required for this event

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Participants access the annual theme on the TSA website under *Themes & Problems*.

D. Participants design, build, and test their robots.

PRELIMINARY ROUND

A. Students will arrive at the assigned place and time to:

   1. Set up their pit area
   2. Set up their robot
   3. Sign up for a robot challenge time

B. Entries are reviewed by judges to determine safety.

C. Safe robots will be given an opportunity to compete in the challenge.

D. The top twelve (12) preliminary scoring teams will move to the semifinal round.

E. A list of twelve (12) semifinalist teams (in random order) will be posted.

SEMIFINAL ROUND

A. Participants report at the time and place stated in the conference program to sign up for an interview time.

B. Participants report at the assigned time and place for the ten (10) minute interview.

C. Robot challenge and interview scores are combined to determine the final standings.

D. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. Participants must check for the current year's design challenge specifications on the TSA website under *Themes & Problems*.

Robotics Pit and Safety Procedures

A. Event coordinator will provide a designated area for teams to work on and prepare their robots for the challenge.

B. Teams must bring, to the pit area, all necessary equipment needed for the challenge, such as robot, parts, chargers, tools, computers etc.

C. Battery Information:

   1. Allowed: Commercially available battery packs of any chemistry, provided they are sealed and non-spillable.
   2. Voltage: The maximum nominal voltage may not exceed twelve (12) volts.

3. Safety: Batteries must be securely mounted within the robot's structure and protected from physical damage.
4. Prohibited: Wet-cell / flooded lead-acid batteries, or any battery pack with visible damage, swelling, or leaks. Lithium-based batteries with soft, flexible, or pouch-style casings are prohibited. All lithium batteries must be enclosed in a commercially manufactured hard plastic or metal protective shell.

D. A judge will inspect the robot to ensure safe operation and that the robot meets specifications.

E. Tools that emit material or sparks are not allowed. Example: Dremel tools

F. Eye protection must be worn in both the pit area and competition area by all team members for the duration of the event.

G. There are no restrictions on structural materials, provided they do not pose a safety risk to participants, the venue, or other robots. Hazardous materials (such as lead, glass, or combustible chemicals) are strictly prohibited.

Robot Specifications

A. Robot must be built from standard, off-the-shelf electronic components. You may use a kit, but it must be a system that allows users to modify the hardware and rewrite the software such as Vex or Tetrix.

B. Robots may be constructed using any safe material, including metal, wood, plastic, or fabric. Commercial parts are allowed and may be modified.

C. Robots must be powered by a single battery pack.

D. Robots must navigate a course with potentially varying surfaces. Teams may utilize any mobility method, including legs, tracks, or any wheel configuration (standard, omni, etc.), provided the robot can move effectively across the terrain.

E. Robots must fit within a 16" x 16" x 16" starting volume. Once the challenge begins, robots may expand beyond these dimensions. The robot must consist of one single entity. Sub-assemblies that split from the main structure, secondary mini-bots, or detachable units (tethered or untethered) are strictly prohibited.

F. A camera may be incorporated into the design if needed to help complete the assigned task.

SEMIFINAL ROUND

A. Participants report at the time and place stated in the conference program to sign up for an interview time.

B. Participants report at the assigned time and place for the ten (10) minute interview.

C. A minimum of two (2) and a maximum of six (6) team members may participate in the interview.

EVALUATION

PRELIMINARY ROUND

A. Robot challenge

SEMIFINAL ROUND

A. Interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Use of Generative Artificial Intelligence (GenAI) is NOT permitted for this event.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability.

Participant/Team ID# _____

ROBOTICS

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Robot

☐ ENTRY NOT EVALUATED

PRELIMINARY ROBOT CHALLENGE (60 points)					
1st: 60 Points	**2nd:** 58 Points	**3rd:** 56 Points	**4th**: 54 Points	**5th:** 52 Points	**6th:** 50 Points
7th: 48 Points	**8th:** 46 Points	**9th:** 44 Points	**10th:** 42 Points	**11th:** 40 Points	**12th:** 38 Points
					Entry evaluated, but did not place in top 12: 1 Point
					PRELIMINARY ROBOT CHALLENGE SUBTOTAL (60 points)

Rules violations (a deduction of 20% of the total possible points in the semifinalist section) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (60 points)

SEMIFINAL INTERVIEW (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Knowledge (X2)	Participants struggle to explain core technical concepts or design choices; answers to questions are vague or off-topic.	Participants accurately explain the basic operation and design choices of their project; answers demonstrate basic technical comprehension.	Participants demonstrate deep understanding of all technical systems, coding logic, and design choices for their robot; questions are answered with specific details that show team members clearly understand all aspects of their project.	
Articulation and Team Participation (X2)	Presentation is disorganized, hard to follow, or illogical; team relies on filler words, struggles to explain ideas, or presentation is dominated by a single speaker.	Presentation is generally clear and organized; team uses basic technical language, follows a logical sequence, and shows coordinated participation.	Presentation is concise, professional, and logically structured; team uses precise engineering terminology, speaks confidently, and demonstrates seamless, balanced participation among all members.	
			SEMIFINAL INTERVIEW SUBTOTAL (40 points)	

Rules violations (a deduction of 20% of the total possible points in the semifinalist section) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
SEMIFINAL SUBTOTAL (40 points)	
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (100 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ROBOTICS
EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Judges, two (2) or more
C. Pit Area Judge/Inspector (1)
D. Assistants, two (2)

MATERIALS

A. Coordinator's packet and box, containing:
    1. Event guidelines, one (1) copy for the coordinator and for each judge
    2. TSA Event Coordinator Report
    3. Stick-on labels for entries, as needed
B. Scoring sheets
C. Semifinalist interview time slot sheet
D. Interview questions
E. Robotics Competition Field
F. Competition materials based on theme
G. Table with power for the Pit Area
H. Table for inspection and tabulation
I. Stanchions or caution tape to separate spectators and participants
J. Safety glasses and vests for judges and coordinators
K. Fire extinguisher and fire blankets

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough personnel have been scheduled.
D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Participants report to the time and place stated in the conference program and check in and sign up for setup/testing time
B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
C. In order to compete, participants must be on the entry list or must have approval of the CRC.
D. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.
E. Secure the entries in the designated area.

PRELIMINARY ROUND

A. Assist judges with the check in/setup of pit area.
B. Assist judges with the challenge.
C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
    1. To deduct twenty percent (20%) of the total possible points in this round
    2. To disqualify the entry

    The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

D. Begin the challenge at the scheduled time.
    1. Every robot that is compliant with rules should have the opportunity to be tested.
    2. Public viewing is allowed.
    3. Announce at starting time: The challenges must be completed in the allotted ten (10)-minute window. If necessary, repairs or adjustments to the robot or its components can be made, but the clock will not stop and no additional time will be awarded to complete the challenges.
E. Position judges to be able to view the challenge.
F. If a time is not properly recorded, a rerun may be ordered at the discretion of the event coordinator.
G. Confirm judges have recorded scores and times.
H. Testing of the robot times will determine the top twelve (12) results.
I. Create the semifinalist sign-up sheet.

SEMIFINAL ROUND

A. Post the top twelve (12) teams with interview times.
B. Teams report to the designated area posted time for a ten (10)-minute interview.
C. Conduct interviews with the qualifying top twelve (12) teams.
D. Begin the interviews at the scheduled time.
E. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
F. If necessary, manage the security and removal of materials from the event area.